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Original Research Article

Biofilm Production and its Role in Antimicrobial Resistance: A Cross-Sectional Study of Clinical Isolates

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ABSTRACT

Background: Biofilm formation is an important virulence factor that contributes to antimicrobial resistance among clinical bacterial isolates. However, the relationship between biofilm production and antimicrobial resistance remains variable across different healthcare settings. This study also aimed to determine the prevalence of biofilm production and evaluate its association with antimicrobial resistance among clinical bacterial isolates in a tertiary care hospital.

Material and Methods: This prospective cross-sectional study was conducted over a period of 6 months in the Department of Microbiology, GMERS Medical College and Hospital, Porbandar. A total of 60 non-duplicate clinical isolates obtained from various specimens including urine, pus, blood, sputum, and device-associated samples were included. Bacterial identification was performed using standard microbiological techniques. Biofilm production was detected using Congo Red Agar (CRA) method and Tube adherence method. Antimicrobial susceptibility testing was performed using the Kirby-Bauer disk diffusion method as per CLSI M100Ed36E guidelines. The association between biofilm production and antimicrobial resistance was analyzed using appropriate statistical tests.

Results: Out of 60 isolates, 31 were identified as biofilm producers while 29 were non-biofilm producers. The predominant organism was E. coli (32), followed by Staphylococcus spp. (15), Pseudomonas (5), Klebsiella (4), and Enterococcus (4). Among 31 biofilm-producing isolates, 71.0% exhibited multidrug resistance.

Conclusion: Biofilm production was observed in a substantial proportion of clinical isolates. A higher rate of multidrug resistance was noted among biofilm-producing isolates, indicating an association between biofilm formation and antimicrobial resistance. However, this relationship may vary depending on the bacterial species.

Keywords: Biofilm, Antimicrobial resistance, Multidrug resistance, Congo red agar, Tube method, Clinical isolates.

INTRODUCTION

Biofilm formation is an important virulence factor in many bacterial pathogens, helping them survive in adverse environments and contributing to evasion of host immune defenses. In biofilms, bacteria are surrounded by an extracellular matrix, which markedly increases their resistance to antimicrobial agents compared with their planktonic forms.[1]

Biofilm-associated resistance leads to persistent infections, higher morbidity, and treatment failure, making it a significant challenge in clinical practice.[2,3]

Antimicrobial resistance (AMR) has emerged as a global health concern, mainly due to the overuse and misuse of antibiotics. Biofilm formation is one of the important mechanisms contributing to resistance, by limiting antibiotic penetration, changing the local environment, and activating resistance genes.[1,3] However, the relationship between biofilm production and antimicrobial resistance is complex and varies among different bacterial species.

Several studies have demonstrated a positive association between biofilm formation and multidrug resistance (MDR). Studies on Escherichia coli and Pseudomonas aeruginosa have reported that strong biofilm producers are more likely to exhibit higher levels of antimicrobial resistance.[4] Similarly, biofilm formation has been shown to increase resistance in organisms such as Klebsiella pneumoniae and Acinetobacter baumannii, complicating the outcome of treatment.[6,7]

In contrast, some studies have reported that the relatio nship between biofilm formation and antimicrobial resistance varies between organisms. A multicentric study on Gram-negative bacteria found no significant overall association with multidrug resistance, although correlations were observed with certain antibiotics. [8] Similarly, studies on Staphylococcus aureus have shown that biofilm formation is not always directly linked to methicillin resistance, suggesting that additional factors also affect resistance patterns.[9,10]

Given these conflicting findings, the relationship between biofilm formation and antimicrobial resistance appears to be context-dependent, varying with bacterial species, site of infection, and local antibiotic use. Therefore, studying this association in different clinical settings is important to better understand infection patterns and guide treatment decisions.

The present study was conducted to determine the prevalence of biofilm production and to assess its association with antimicrobial resistance among clinical bacterial isolates from a tertiary care hospital.

MATERIAL AND METHODS

A total of 60 non-duplicate clinical bacterial isolates were included in the study. These isolates were obtained from different clinical specimens such as urine, pus, blood, and sputum. Only clinically significant isolates were included, and duplicate isolates from the same patient were excluded.

Bacterial Isolation & Identification

- Inoculation on Blood agar, MacConkey agar, and Nutrient agar.
- Incubation at 37°C for 24–48 hours.
- Identification by Gram staining and standard biochemical tests.

Biofilm Detection Methods

1. Congo Red Agar (CRA) method: Black, dry crystalline colonies = Biofilm producers. CRA is a qualitative assay for detecting biofilm formation. Bacterial isolates are inoculated onto Congo red agar plates (Brain-heart infusion agar supplemented with 5% sucrose and Congo red dye). After incubation at 37°C for 24–48 hours, biofilm producers appear as black, dry, crystalline colonies, whereas non-biofilm formers appear as red or pink colonies.

2. Tube Adherence Method (Christensen's method): Safranin-stained film lining the tube wall = Positive. This is a simple phenotypic test. Isolates are inoculated into glass test tubes containing tryptic soy broth supplemented with 1% glucose and incubated at 37°C for 24–48 hours. After incubation, tubes are decanted, washed with phosphate-buffered saline, stained with 0.1% safranin or gentian violet, and rinsed with distilled water. A visible stained film lining the inner wall and bottom of the tube indicates biofilm production, while absence of film indicates non-producers.

Antimicrobial Susceptibility Testing (AST)

- Performed by Disk-diffusion (Kirby-Bauer) method and/or MIC methods as per CLSI M100Ed36E.
- Antibiotics tested included commonly used agents against Gram-negative bacilli (Amikacin, Gentamicin, cephalosporins, piperacillin-tazobactam, carbapenems, fluoroquinolones, nitrofurantoin, cotrimoxazole, cefepime, ceftazidime, ceftriaxone, and aztreonam) and Gram-positive cocci (Penicillin, Ampicillin, Vancomycin, Cefoxitin, Linezolid, Erythromycin, Tetracycline).
- Quality control was performed using reference ATCC strains following CLSI-recommended QC ranges and frequencies.

RESULTS

A total of 60 non-duplicate clinical isolates were included in the study. Among these, 31 (51.7%) were identified as biofilm producers, while 29 (48.3%) were non-biofilm producers (Table 1).

The predominant organism isolated was Escherichia coli (32; 53.3%), followed by Staphylococcus spp. (15; 25.0%), Pseudomonas spp. (5; 8.3%), Klebsiella spp. (4; 6.7%), and Enterococcus spp. (4; 6.7%) (Chart 1).

Among the 31 biofilm-producing isolates, 22 (71.0%) were multidrug resistant (MDR), whereas among the 29 non-biofilm producers, only 12 (41.4%) were MDR, indicating a higher proportion of MDR among biofilm-producing isolates (Chart 2).

The antibiotic resistance pattern of Gram-negative bacilli showed higher resistance in biofilm-producing organisms compared to non-biofilm-producing ones. In biofilm-producing E. coli, the highest resistance was seen with Ampicillin (66.7%) and Levofloxacin (66.7%), followed by Cefotaxime (60%), Ceftriaxone (60%), and Tetracycline (60%). In non-biofilm-producing E. coli, resistance was lower: Ampicillin (41.2%) and Levofloxacin (29.4%), followed by Cefotaxime (17.6%), Ceftriaxone (17.7%), and Tetracycline (35.3%). The most sensitive agents in both biofilm-producing and non-biofilm-producing E. coli were carbapenems (Meropenem and Imipenem). The most sensitive agents in both biofilm-producing and non-biofilm-producing Pseudomonas spp. were carbapenems, fluoroquinolones, and piperacillin-tazobactam (Table 2).

Similarly, the antibiotic resistance pattern of Gram-positive cocci showed higher resistance in biofilm-producing organisms compared to non-biofilm-producing ones. In biofilm-producing MRSA, the highest resistance was to Gentamicin (66.7%), followed by Ciprofloxacin, Levofloxacin, Clindamycin, Azithromycin, Erythromycin, Sulfamethoxazole-trimethoprim, and Tetracycline (33.3% each). In non-biofilm-producing MRSA, resistance was: Ciprofloxacin (60%), Gentamicin (40%), followed by Erythromycin, Azithromycin, and Levofloxacin (20% each). The most sensitive agents in both biofilm-producing and non-biofilm-producing Staphylococcus aureus were Linezolid, Vancomycin, Tetracycline, Doxycycline, and Clindamycin. The most sensitive agents in both biofilm-producing and non-biofilm-producing Enterococcus species were Linezolid, Vancomycin, Tetracycline, and Doxycycline (Table 3).

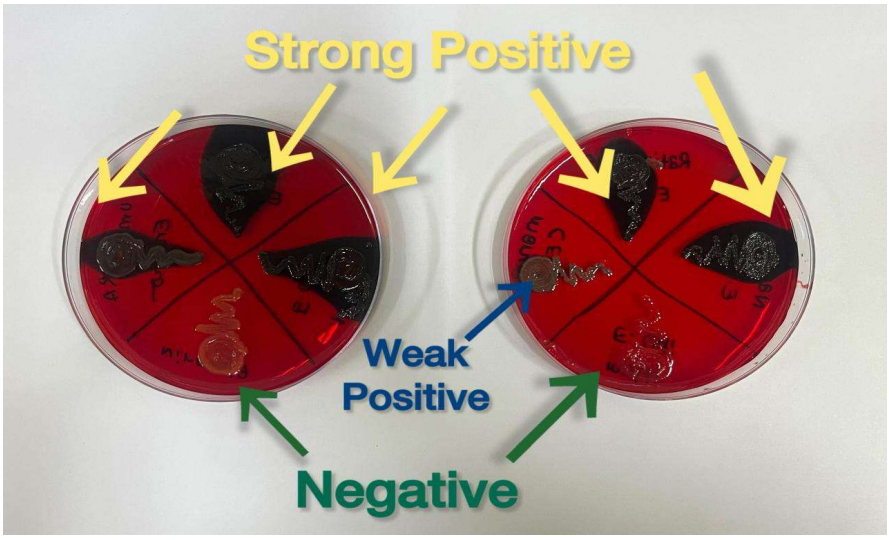


Figure - 1: Congo Red Agar method showing crystalline black & dry colonies indicating biofilm-producing isolates; red or pink colonies indicate non-biofilm-producing isolates.

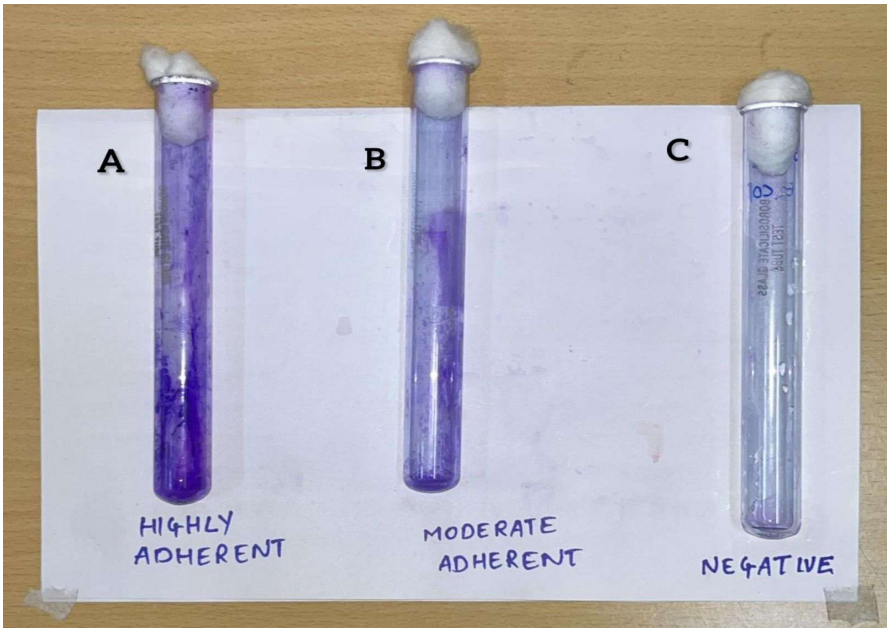


Figure - 2: Tube method showing A & B – wall and bottom of tubes with visible violet colour, indicating biofilm-producing isolates; C – no visible violet colour, indicating a non-biofilm-producing isolate.

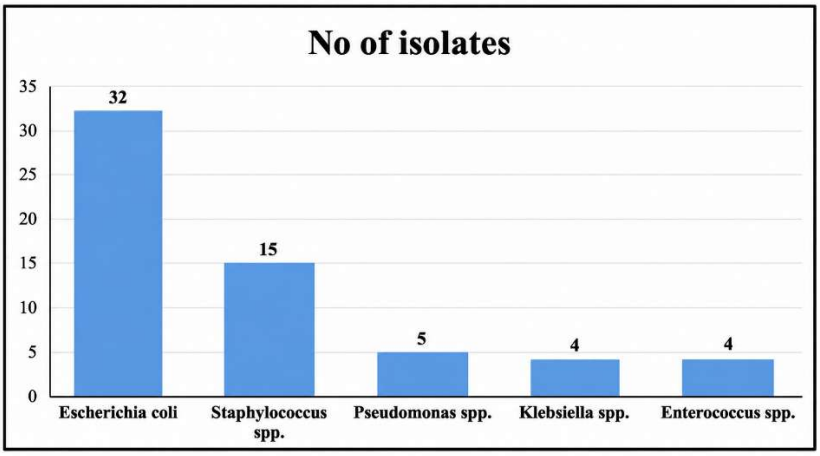


Chart -1: Organism isolated from clinical samples (n=60).

Table – 1: Distribution of Clinical Isolates by Biofilm Producer and Non-Producer Status

Organism	Biofilm Producer	Biofilm Non-Producer
Escherichia coli (n=32)	**15 (46.9%)**	**17 (53.1%)**
Pseudomonas spp. (n=5)	**2 (40%)**	**3 (60%)**
Klebsiella spp. (n=4)	**4 (100%)**	**0**
Staphylococcus spp. (n=15)	**7 (46.7%)**	**8 (53.3%)**
Enterococcus spp. (n=4)	**3 (75%)**	**1 (25%)**
Total (n=60)	**31 (51.7%)**	**29 (48.3%)**

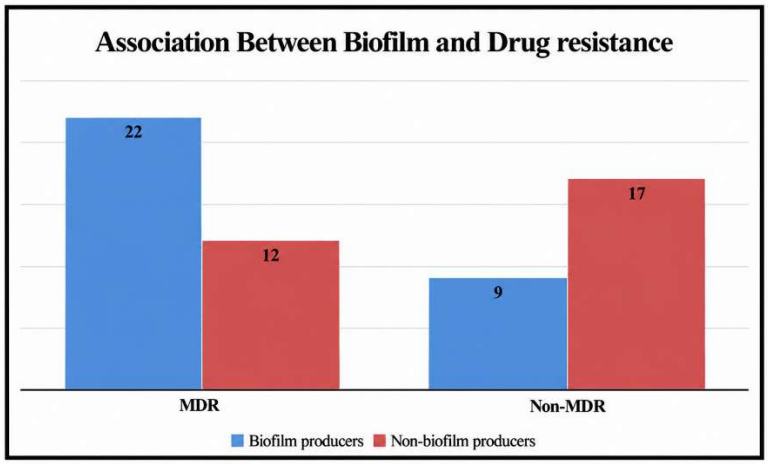


Chart 2: Association between biofilm production and drug resistance (MDR and Non-MDR) among organisms isolated from samples (n=60).

Table – 2: Antibiotic Resistance Pattern of Gram-Negative Bacilli, Biofilm Producer vs. Non-Producer

Antibiotics	E. coli (n=15) Biofilm producer	Klebsiella spp. (n=4) Biofilm producer	Pseudomonas spp. (n=2) Biofilm producer	E. coli (n=17) Biofilm non-producer	Pseudomonas spp. (n=3) Biofilm non-producer
Sulfamethoxazole-trimethoprim	46.7%	0%	-	41.2%	-
Gentamicin	33.3%	25%	-	5.9%	-
Tetracycline	60%	50%	-	35.3%	-
Levofloxacin	66.7%	25%	50%	29.4%	0%
Ciprofloxacin	33.3%	25%	50%	41.2%	0%
Amikacin	40%	25%	50%	41.2%	0%
Meropenem	6.7%	25%	50%	0%	0%
Imipenem	6.7%	25%	50%	0%	0%
Ampicillin	66.7%	100%	-	41.2%	-
Ceftriaxone	60%	25%	-	17.7%	-
Amoxiclav	73.3%	100%	-	41.2%	-
Ceftazidime	53.3%	50%	50%	23.5%	33.3%
Cefepime	46.7%	75%	50%	17.6%	33.3%
Piperacillin-tazobactam	53.3%	50%	50%	29.4%	0%
Cefotaxime	60%	50%	-	17.6%	-

Table – 3: Antibiotic Resistance Pattern of Gram-Positive Cocci, Biofilm Producer vs. Non-Producer

Antibiotics	MSSA (n=4) Biofilm producer	MRSA (n=3) Biofilm producer	Enterococcus spp. (n=3) Biofilm producer	MSSA (n=3) Biofilm non-producer	MRSA (n=5) Biofilm non-producer	Enterococcus sp. (n=1) Biofilm non-producer
Penicillin	50%	-	66.7%	0%	-	0%
Cefoxitin	0%	-	-	0%	-	-
Chloramphenicol	0%	0%	-	0%	0%	-
Sulfamethoxazole-trimethoprim	25%	33.3%	-	0%	0%	-
Erythromycin	0%	33.3%	-	0%	20%	-
Azithromycin	0%	33.3%	-	0%	20%	-
Clindamycin	0%	33.3%	-	0%	0%	-
Gentamicin	0%	66.7%	-	66.7%	40%	-
High Level Gentamicin	-	-	66.7%	-	-	0%
Tetracycline	0%	0%	0%	0%	0%	0%
Doxycycline	0%	0%	0%	0%	0%	0%
Linezolid	0%	0%	0%	0%	0%	0%
Levofloxacin	0%	33.3%	66.7%	0%	20%	0%
Ciprofloxacin	0%	33.3%	100%	0%	60%	0%
Vancomycin	0%	0%	0%	0%	0%	0%
Ampicillin	-	-	66.7%	-	-	0%

DISCUSSION

This study was designed to evaluate the prevalence of biofilm formation and its association with antibiotic resistance among clinical bacterial isolates from a tertiary care hospital. Sixty isolates were examined, and 31 (51.7%) were found to be biofilm producers. The most frequently isolated organism was *Escherichia coli* (53.3%), followed by

Staphylococcus spp. (25.0%), *Pseudomonas* spp. (8.3%), *Klebsiella* spp. (6.7%), and *Enterococcus* spp. (6.7%). Notably, all four *Klebsiella* spp. isolates were biofilm producers, whereas biofilm production varied among the other bacterial species. These findings are consistent with prior observations that biofilm formation is a common phenomenon among clinical infections and that it is a key virulence component.[2,3]

The prevalence of 51.7% is within the vast range recorded from tertiary care centers in India. Isolates from orthopedic implant infections in a tertiary care hospital at Surat, Gujarat revealed less prevalence (37.7%).[16] Uropathogens in Rohtak, Haryana showed 45.6%.[17] Biofilm production was reported in 55% of *Staphylococcus aureus* isolates in a study from Tripura, with a higher prevalence among MRSA (76.6%) than MSSA (38.8%).[18] Similarly, a study conducted at Dr. D.Y. Patil Hospital and Research Centre reported biofilm production in 59.23% of clinical staphylococcal isolates.[19] A study from Chennai also reported biofilm formation among Enterococcus isolates.[15] In a study from Puducherry, all 106 uropathogenic *Escherichia coli* isolates were classified as weak, moderate or strong biofilm producers, and biofilm-production category was significantly associated with multidrug resistance.[20] Differences in study populations, biofilm detection methods and classification criteria should be considered when comparing reported biofilm prevalence across studies. The staphylococcal study reported a higher proportion of biofilm-positive isolates using the tube adherence method (50%) than the Congo Red Agar method (31.5%), highlighting the influence of the detection method on reported biofilm prevalence.[19] Prevalence figures from different centers are thus not directly comparable.

Biofilm formation confers a major survival benefit for bacteria by encapsulating them within a self-produced extracellular matrix that restricts antibiotic penetration and facilitates persistence in the host. Bacteria in biofilms are far more resistant than planktonic bacteria due to low metabolic activity, limited diffusion of antimicrobials and activation of resistance genes.[1,3] These features contribute to persistent, recurrent infections and greater therapeutic failure representing a serious issue in clinical settings.

In the present investigation, biofilm producers had a greater proportion of multidrug resistance (MDR) compared to non-biofilm producers (71.0% vs. 41.4%), demonstrating a positive connection between biofilm formation and antimicrobial resistance. These findings are consistent with reports from other Indian tertiary-care settings. In Rohtak, all biofilm-producing uropathogens were reported to be multidrug-resistant (MDR).[17] In Puducherry, biofilm-production category was significantly associated with MDR among uropathogenic *E. coli*.[20] Similarly, the Surat study reported greater antimicrobial resistance among biofilm-producing isolates.[16] However, a considerable proportion of non-biofilm-producing isolates in the present study (41.4%) were also MDR.

The antibiotic resistance patterns are presented in Tables 2 and 3. Overall, the biofilm-producing Gram-negative bacilli were more resistant, although not for all the agents. Resistance to ampicillin (66.7%), levofloxacin (66.7%), cefotaxime (60%), ceftriaxone (60%), and amoxiclav (73.3%) was higher among biofilm-producing *E. coli* isolates than among non-biofilm-producing isolates, while resistance to ciprofloxacin (33.3% vs 41.2%) and amikacin (40% vs 41.2%) was not higher. These observations are consistent with previous reports on E. coli and Pseudomonas aeruginosa where strong biofilm producers exhibited higher resistance rates,[4,5] and with the high resistance to cephalosporins reported in biofilm-forming E. coli from Puducherry.[20] Carbapenems (meropenem and imipenem) retained good activity against E. coli in both groups. These findings should be interpreted in the context of local resistance patterns, as the Surat study reported meropenem susceptibility in only 47% of the isolates.[16] This difference highlights the variation in antimicrobial susceptibility patterns across healthcare settings.

All the four Klebsiella spp. isolates in the present investigation were biofilm producers. They showed high rates of resistance to ampicillin (100%), amoxiclav (100%), cefepime (75%) and carbapenem resistance (25%). Similar biofilm development and resistance was previously described in Klebsiella pneumoniae involving 100% ampicillin resistance in ESBL- and KPC-producing isolates.[6,13] However, with only four isolates, this cannot be extrapolated to imply that all Klebsiella isolates generate biofilm; in the Surat study, Klebsiella and Pseudomonas isolates comprised

both biofilm producers and non-producers.[16] The two biofilm-producing isolates of Pseudomonas demonstrated 50% resistance to the drugs tested. However, the small number of *Pseudomonas* isolates limits meaningful comparisons with other studies.

However, the association between biofilm formation and antimicrobial resistance varied across bacterial species and antibiotics. Among Gram-positive cocci (Table 3), biofilm-producing MRSA isolates demonstrated stronger resistance than non-biofilm-producing MRSA to gentamicin (66.7% vs 40%) and levofloxacin (33.3% vs 20%), but ciprofloxacin resistance was higher in non-producers (60% vs 33.3%). Resistance to gentamicin in MSSA was found only in non-producers (66.7% versus 0%). Tripura showed a similar mixed pattern, with higher ciprofloxacin resistance in biofilm producing S. aureus, while gentamicin, tetracycline and penicillin resistance was higher in non-producers.[18] Linezolid and vancomycin were uniformly active against both biofilm producing and non-producing staphylococci, as expected from reports that biofilm formation does not necessarily abrogate susceptibility to these agents,[9],[14] and from their full activity against S. aureus in Surat[16] and against Gram positive biofilm producers in Rohtak.[17]

Among the three biofilm-producing *Enterococcus* isolates, resistance was observed to ciprofloxacin (100%), high-level gentamicin (66.7%), ampicillin (66.7%), and levofloxacin (66.7%). Linezolid, vancomycin, tetracycline and doxycycline were 100% sensitive in both biofilm and non-biofilm producers. A study from Chennai also reported that antimicrobial resistance was not consistently associated with biofilm production; *Enterococcus faecalis* isolates showed greater antimicrobial resistance despite lower biofilm formation.[15] However, in the study vancomycin resistance was noted in 22% of isolates and linezolid resistance in 9% of isolates, contrary to our results.[15] This discrepancy may be related to the modest number of Enterococcus isolates in our investigation (n=4), the species distribution and the local resistance trends.

These disparities across different organisms and studies may be due to different bacterial species, sample size, clinical situation and methodologies used for detection of biofilm. Biofilm formation is a complex process involving genetic determinants, environmental variables, host-related factors and history of antibiotic exposure, all of which can contribute to the observed heterogeneity in resistance patterns across diverse environments.

Overall, the findings suggest that biofilm formation may contribute to antimicrobial resistance, although the association varies across bacterial species and antibiotics. The observed differences in resistance patterns highlight the importance of organism-specific investigations of biofilm-associated resistance. Multidrug-resistant biofilm-producing isolates may pose challenges for clinical management, underscoring the importance of appropriate antimicrobial susceptibility testing and judicious antibiotic use. Although carbapenems, linezolid and vancomycin retained activity against the relevant isolates in this study, their continued effectiveness should be supported by appropriate antimicrobial stewardship. Routine biofilm detection, alongside antimicrobial susceptibility testing, may provide complementary information for the assessment of infections caused by biofilm-forming organisms.

CONCLUSIONS

Biofilm production was observed in 51.7% of clinical isolates in this study. A significantly higher rate of multidrug resistance (71.0%) was noted among biofilm-producing isolates compared to non-biofilm producers (41.4%), indicating a positive association between biofilm formation and antimicrobial resistance. Antibiotic resistance profiling revealed that biofilm-producing Gram-negative bacilli, particularly E. coli and Klebsiella spp., showed higher resistance to fluoroquinolones, cephalosporins, and penicillin compared to their non-biofilm-producing counterparts, while carbapenems retained activity. Among Gram-positive cocci, MRSA biofilm producers showed higher fluoroquinolone and aminoglycoside resistance; however, Linezolid, Vancomycin, Tetracycline, and

Doxycycline remained active against all Staphylococcus spp. and Enterococcus spp. isolates. These findings suggest that biofilm production contributes to increased antimicrobial resistance, though the pattern is organism- and antibiotic-specific. Routine detection of biofilm formation in clinical microbiology laboratories, combined with species-level antibiotic susceptibility testing, may serve as a valuable tool for predicting resistance profiles and guiding appropriate therapeutic decisions in biofilm-associated infections.

LIMITATIONS

The present study has certain limitations. The sample size was relatively small and limited to a single tertiary care center, which may affect the generalizability of the findings. A mixed group of bacterial isolates was included, and organism-wise analysis of the association between biofilm formation and antimicrobial resistance was not carried out, which could have provided more detailed insights.

Biofilm detection was performed using phenotypic methods (Congo Red Agar and tube adherence method), which are simple and cost-effective but may lack the sensitivity and reproducibility of more advanced quantitative methods such as microtiter plate assays.[11] in addition, molecular analysis of biofilm-related genes and resistance mechanisms was not performed.[12]

AUTHOR CONTRIBUTIONS

Ria Kotecha: Conceived the study design, oversaw laboratory procedures, and contributed to data interpretation and manuscript preparation.

Chandani Surani: Provided clinical correlation, and contributed to manuscript review and clinical interpretation.

Ahuti Pandya: Conducted statistical analysis, contributed to data interpretation, performed literature review, and drafted and revised the manuscript.

Payal Odedara: Performed microbiological identification and contributed to laboratory procedures.

REFERENCES

1. Liu HY, Prentice EL, Webber MA. Mechanisms of antimicrobial resistance in biofilms. NPJ Antimicrob Resist. 2024;2:27. doi:10.1038/s44259-024-00046-3

2. Tan C, Wang J, Wu A, Li C. Mapping the global landscape of biofilm-associated antimicrobial resistance (1992–2025). Biofilm. 2026;11:100358. doi:10.1016/j.bioflm.2026.100358.

3. Mbaraka A, Meena RR, Menghani E, Verma N. Targeting biofilm-driven antibiotic resistance: emerging mechanisms and next-generation therapeutic interventions. Front Microbiol. 2026;17:1823476. doi:10.3389/fmicb.2026.1823476.

4. Qian W, Li X, Yang M, Liu C, Kong Y, Li Y, et al. Relationship between antibiotic resistance, biofilm formation, and biofilm-specific resistance in *Escherichia coli* isolates from Ningbo, China. Infect Drug Resist. 2022;15:2865-78. doi:10.2147/IDR.S363652.

5. Karami P, Khaledi A, Yousefi Mashouf R, Yaghoobi MH, Karami M, Dastan D, et al. The correlation between biofilm formation capability and antibiotic resistance pattern in *Pseudomonas aeruginosa*. Gene Rep. 2020;18:100561. doi:10.1016/j.genrep.2019.100561.

6. Li L, Gao X, Li M, Liu Y, Ma J, Wang X, et al. Relationship between biofilm formation and antibiotic resistance of *Klebsiella pneumoniae* and updates on antibiofilm therapeutic strategies. Front Cell Infect Microbiol. 2024;14:1324895. doi:10.3389/fcimb.2024.1324895.

7. Javadi K, Ghaemian P, Baziboron M, Pournajaf A. Investigating the link between biofilm formation and antibiotic resistance in clinical isolates of *Acinetobacter baumannii*. Int J Microbiol. 2025;2025:1009049. doi:10.1155/ijm/1009049.

8. Cepas V, López Y, Muñoz E, Rolo D, Ardanuy C, Martí S, et al. Relationship between biofilm formation and antimicrobial resistance in Gram-negative bacteria. Microb Drug Resist. 2019;25(1):72-9. doi:10.1089/mdr.2018.0027.

9. Tahaei SAS, Stájer A, Barrak I, Ostorházi E, Szabó D, Gajdács M. Correlation between biofilm-formation and the antibiotic resistant phenotype in *Staphylococcus aureus* isolates: a laboratory-based study in Hungary and a review of the literature. Infect Drug Resist. 2021;14:1155-68. doi:10.2147/IDR.S303992.

10. Dančová N, Király J, Hajdučková V, Hudecová P, Hisirová S, Nagyová M, et al. Characterization and antimicrobial resistance profiles of biofilm forming strains of *Staphylococcus aureus* isolated from skin lesions. Microorganisms. 2025;13(11):2449. doi:10.3390/microorganisms13112449.

11. Mukashema H, Yadufashije C, Tuyishimire A, Imurinde Y, Niyonkuru BR, Muhimpundu L, et al. Profile of biofilm formation and antimicrobial susceptibility patterns of *Escherichia coli* isolated from adult patients presenting with urinary tract infections at Ruhengeri Level Two Teaching Hospital, Rwanda. Infect Drug Resist. 2026;19:559490. doi:10.2147/IDR.S559490.

12. Mokari S, YousefiMashouf R, Karami P, Taheri M. Genetic diversity, antimicrobial resistance, and biofilm-associated virulence in clinical *Enterococcus faecalis* isolates from Hamedan, Iran. BMC Infect Dis. 2026;26:544. doi:10.1186/s12879-026-12768-y.

13. Nazari M, Hemmati J, Asghari B. Comprehensive analysis of virulence genes, antibiotic resistance, biofilm formation, and sequence types in clinical isolates of *Klebsiella pneumoniae*. Can J Infect Dis Med Microbiol. 2024;2024:1403019. doi:10.1155/cjid/1403019.

14. Kaushik A, Kest H, Sood M, Steussy BW, Thieman C, Gupta S. Biofilm producing methicillin-resistant *Staphylococcus aureus* (MRSA) infections in humans: clinical implications and management. Pathogens. 2024;13(1):76. doi:10.3390/pathogens13010076.

15. Nair P, Sankar S, Neelusree P. Study on biofilm formation among Enterococcus isolates and association with their antibiotic resistance patterns. Cureus. 2024;16(2):e53594. doi:10.7759/cureus.53594.

16. Sarkar K, Mullan S, Menon H. Exploring drug resistance: microbial profiles, antibiotic sensitivity, and biofilm development in orthopedic implant infections. Cureus. 2024;16(10):e70938. doi:10.7759/cureus.70938.

17. Panda PS, Chaudhary U, Dube SK. Study of biofilm production and antimicrobial sensitivity pattern of uropathogens in a tertiary care hospital in North India. Int J Community Med Public Health. 2016;3(9):2421-6. doi:10.18203/2394-6040.ijcmph20163049.

18. Bhattacharya S, Bir R, Majumdar T. Evaluation of multidrug resistant *Staphylococcus aureus* and their association with biofilm production in a tertiary care hospital, Tripura, Northeast India. J Clin Diagn Res. 2015;9(9):DC01-4. doi:10.7860/JCDR/2015/13965.6417.

19. Singh S, Asthana AK, Madan M, Ghosh SJ. Study of biofilm formation among clinical staphylococcal isolates. Natl J Lab Med. 2015;4(4):24-7. doi:10.7860/NJLM/2015/13975:2062.

20. Shebeena S, Ragunathan L, Kannaiyan K, Sanjeevi T, Subramanian P, Sasi AC, et al. Phenotypic, genotypic characterization and antimicrobial resistance profiling of uropathogenic *Escherichia coli* in a tertiary care hospital, Puducherry, India. Iran J Microbiol. 2025;17(3):366-75. doi:10.18502/ijm.v17i3.18818.

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